Dimensionality Extension Plan
Belief-State (Bayesian) Adaptive Fractionation

Internal repository design note

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1 Purpose

This note defines the belief-state extension to the adaptive fractionation solver across two stages of increasing Bayesian fidelity. Both stages retain a 5-coordinate finite-horizon state.

The baseline (current master branch) state is:

$$\text{state}_{\text{current}} = (t,\, o_t,\, D_t). \tag{1}$$

Stage A (implemented) extends the state to include a running Gaussian belief:

$$\text{state}_{\text{A}} = (t,\, o_t,\, D_t,\, \mu_t,\, \sigma_t), \tag{2}$$

where (μ_t, σ_t) tracks a running mean and standard deviation of the patient's PTV–OAR overlap distribution, updated after each observed fraction via Welford's online algorithm. Branch transition probabilities are computed from a Normal predictive distribution $\mathcal{N}(\mu_t, \sigma_t^2)$.

Stage B (NIG extension) retains the same 5-coordinate state but replaces the Normal predictive with the theoretically correct posterior predictive derived from the Normal-Inverse-Gamma (NIG) conjugate model. The key observation enabling this is that the full NIG posterior is a deterministic function of the sufficient statistics $(n, \bar{s}, M_2)$, all of which are already encoded in the Stage A state (μ_t, σ_t, t) together with the fixed population prior. No additional state dimensions are required: Stage B is implemented by changing how branch-probability tables are computed, not by expanding the state.

2 Scope and Non-Goals

2.1 In Scope

- Replace the Normal branch-probability tables with NIG posterior-predictive (Student-t) tables, indexed by fraction number.
- Use the prior-corrected posterior mean $\mu_n = (\kappa_0\mu_0 + n\bar{s})/(\kappa_0 + n)$ as the predictive centre rather than the raw sample mean.
- Update planning-time initialisation to use the full NIG posterior predictive rather than a Normal approximation.
- Estimate and introduce the two additional NIG prior hyperparameters μ_0 (prior mean) and κ_0 (prior precision weight) from population-level estimation on the patient cohort, stored alongside the existing `DEFAULT_ALPHA` and `DEFAULT_BETA` constants.

- Preserve Welford's belief-update rule unchanged: it is equivalent to updating the NIG sufficient statistics.
- Update the planning-scan termination criterion in `precompute_plan` to use a fixed clinically safe overlap bound rather than a Gaussian percentile heuristic.

2.2 What Remains Unchanged

- The state coordinates $(t, o_t, D_t, \mu_t, \sigma_t)$ and their discretisation grids.
- The belief-update operator: Welford's online algorithm.
- Per-fraction action bounds $d_t \in [d_{\min}, d_{\max}]$ with the same action discretisation by `dose_steps`.
- Feasibility screening: infeasible states are assigned a hard sentinel value using the same convention as Stage A.
- Terminal prescription logic: only $D_F = D_{\text{pres}}$ is feasible at treatment end; terminal underdose and overdose remain hard-infeasible.

2.3 Out of Scope

- Changes to the clinical objective form beyond existing penalty logic.
- UI redesign in `app.py` beyond minimal parameter exposure.

3 Required Mathematical Changes

3.1 Stage A (Implemented): Bellman Recursion

The Stage A recursion is:

$$V_t^{\mathrm{A}}(o, D, \mu, \sigma) = \max_d \Big[-\hat{c}_t(o, d) + \mathbb{E}_{o' \sim \mathcal{N}(\mu, \sigma^2)} V_{t+1}^{\mathrm{A}}\big(o', D + d,\ \mathcal{U}_{\mathrm{W}}(\mu, \sigma, o', t)\big) \Big]. \tag{3}$$

The belief update $\mathcal{U}_{\mathrm{W}}$ is Welford's running-moment increment. Let $n_t = t$ be the number of observations accumulated at fraction t; then:

$$\mu' = \frac{n_t\, \mu + o'}{n_t + 1}, \tag{4}$$

$$(\sigma')^2 = \frac{n_t\, \sigma^2 + (o' - \mu)(o' - \mu')}{n_t + 1}. \tag{5}$$

This is the population-variance (ddof= 0) form of Welford's increment, consistent with the codebase convention and with the denominator convention of the 2025 paper.
Initialisation at planning time. At decision time for fraction f (with f real observations $o_1, \ldots, o_f$): set $\mu_{\text{start}} = \bar{o}_f$ (population mean of observations) and σ_{start} from the MAP estimator (equations 11–13 of the clinical trial paper) applied to all f observations, using prior parameters (α_0, β_0). This gives a prior-regularised σ that is nonzero even at $f = 1$.
Branch representative. For each overlap bin j with edges $(l_j, u_j]$, use the grid centre `volume_space[j]` as the concrete scalar fed into $\mathcal{U}_{\mathrm{W}}$.

3.2 Stage A: Grid Specification

The μ grid uses five non-uniform segments over $[0, 30]$ cc, with finest resolution in the typical-patient range and progressively coarser resolution toward extreme values:

Segment	**Range (cc)**	**Points**	**Step (cc)**
1	[0.00, 1.00]	38	≈ 0.027
2	[1.05, 4.00]	39	≈ 0.078
3	[4.10, 10.00]	42	≈ 0.144
4	[10.20, 16.00]	15	≈ 0.414
5	[16.50, 30.00]	16	≈ 0.900
Total		**150**	

The σ grid uses 11 non-uniform points over $[0.05, 4.5]$ cc, with fine resolution in the range where most patients fall and coarser resolution in the tail:

Segment	**Range (cc)**	**Points**	**Step (cc)**
1	[0.05, 0.70]	5	≈ 0.163
2	[0.80, 1.80]	3	≈ 0.500
3	[2.00, 4.50]	3	≈ 1.250
Total		**11**	

This gives $150 \times 11 = 1{,}650$ belief states. The upper limit of 4.5 cc is set to avoid clipping the most variable patient in the 58-patient cohort (patient 3, MAP $\hat{\sigma} = 4.05$ cc). The fixed overlap state space `_VOLUME_SPACE` covers $[0, 44]$ cc at 441 uniformly spaced points (step 0.1 cc, matching TPS volume discretisation). The 44 cc bound is hardcoded and independent of $\sigma_{\max}$, so the bin width stays exactly 0.1 cc regardless of future σ grid changes; 44 cc comfortably covers $\mu_{\max} + 3\sigma_{\max} = 30 + 3 \times 4.5 = 43.5$ cc and the tail probability beyond 44 cc is negligible for any belief on the grid. Branch probabilities fold left- and right-tail mass into the boundary bins so that probabilities sum to 1 for every belief-grid point. This tail-folding argument for Stage A relies on the Normal predictive being negligibly thin beyond 44 cc. For Stage B, before committing the 44 cc upper bound, sweep all $(t, \mu_{\text{grid}}, \sigma_{\text{grid}})$ triples under the fitted NIG hyperparameters and record the Student-t predictive tail mass above 44 cc. If the maximum folded mass is non-negligible, enlarge `_VOLUME_SPACE` (e.g. to 50 or 60 cc) and re-run the benchmark to measure policy sensitivity; alternatively, document the approximation error explicitly if the 44 cc cap is operationally required.

Numerical invariant. `_VOLUME_SPACE` is fixed and shared across all beliefs and all fractions. Using patient-specific or belief-specific overlap ranges is explicitly prohibited: when a belief's μ is far from the patient's current range, a narrow range assigns nearly all probability mass to a single edge bin, making the DP plan as if all future overlaps equal that edge regardless of the true belief. A fixed wide range avoids this.

3.3 Stage B: NIG Posterior from Sufficient Statistics

The NIG model places a Normal-Inverse-Gamma prior on the unknown overlap-distribution parameters (μ, σ^2):

$$\mu,\, \sigma^2 \;\sim\; \text{NIG}(\mu_0,\, \kappa_0,\, \alpha_0,\, \beta_0).$$

Given n i.i.d. observations with sample mean $\bar{s}$ and sum of squared deviations $M_2 = \sum_{i=1}^{n}(o_i - \bar{s})^2$, the posterior is $\mathrm{NIG}(\mu_n, \kappa_n, \alpha_n, \beta_n)$ with the closed-form updates:

$$\kappa_n = \kappa_0 + n, \tag{6}$$

$$\mu_n = \frac{\kappa_0\,\mu_0 + n\,\bar{s}}{\kappa_n}, \tag{7}$$

$$\alpha_n = \alpha_0 + \tfrac{n}{2}, \tag{8}$$

$$\beta_n = \beta_0 + \tfrac{M_2}{2} + \frac{\kappa_0\,n\,(\bar{s} - \mu_0)^2}{2\,\kappa_n}. \tag{9}$$

Key reduction. The Stage A state encodes all four sufficient statistics:

$n = t$ (absolute 1-based fraction index, tracked as `observation_count = int(fraction_number)`),
$\bar{s} = \mu_t$ (Welford running mean, a state coordinate),
$M_2 = n\,\sigma_t^2$ (recoverable from σ_t and t).

Therefore $(\kappa_n, \mu_n, \alpha_n, \beta_n)$ are computable deterministically at any DP node from (μ_t, σ_t, t) and the fixed prior constants $(\mu_0, \kappa_0, \alpha_0, \beta_0)$. No additional state coordinates are required.
Two practical caveats. First, the grid quantisation of σ_t introduces the same discretisation error as in Stage A and does not degrade Stage B relative to Stage A. Second, the Welford update in `_hypothetical_belief_grid_indices` and `_bellman_expectation_full_grid` applies a floor $\sigma' \geq \sigma_{\min} = 0.05\,\mathrm{cc}$ before snapping to the grid; the effective M_2 entering the β_n computation is therefore $n \times \max(\sigma_t^2, \sigma_{\min}^2)$, not $n \times \sigma_t^2$. This is a minor effect for typical patients but must be matched exactly in the β_n computation to keep Stage B consistent with Stage A's Welford dynamics.

3.4 Stage B: Posterior Predictive Distribution

Given the posterior $\mathrm{NIG}(\mu_n, \kappa_n, \alpha_n, \beta_n)$, the posterior predictive distribution for the next observation is:

$$o_{t+1} \mid (\mu_t,\, \sigma_t,\, t) \;\sim\; \text{Student-t}_{2\alpha_n}\big(\mu_n,\; s_n\big), \tag{10}$$

where the predictive scale is

$$s_n = \sqrt{\frac{\beta_n\,(\kappa_n + 1)}{\alpha_n\,\kappa_n}}, \tag{11}$$

and all quantities on the right-hand side are computed from $(\mu_t, \sigma_t, t, \mu_0, \kappa_0, \alpha_0, \beta_0)$ via equations (6)–(9).
Why Student-t. When both the mean and variance of the overlap distribution are uncertain and marginalised out, the predictive is Student-t rather than Normal. This gives heavier tails, most pronounced at early fractions where α_n is small and the degrees of freedom $2\alpha_n = 2\alpha_0 + t$ are low (near-Cauchy at $t = 1$). Here "near-Cauchy" means a very low-degree-of-freedom Student-t: not literally a Cauchy unless the degrees of freedom equal 1, but close in the practical sense that it is much less concentrated than a Normal and assigns substantially more probability to extreme overlaps far from its centre. As t grows, $2\alpha_n \to \infty$ and the Student-t converges to a Normal; Stage B and Stage A become equivalent in the large-t limit.
Finite-variance constraint. Enforce $\alpha_0^{\mathrm{NIG}} \geq 1$ in the empirical Bayes optimisation, which guarantees degrees of freedom $2\alpha_n \geq 2+t$ at every fraction and therefore a finite-variance predictive throughout

treatment. Note that $\alpha_0 = 1$ is the minimum acceptable value but remains the boundary case: at fraction 1 the predictive is Student-t_3, which has finite variance but still places roughly five times more probability mass beyond $3 \times$ scale than a Normal. The fraction-1 predictive should be plotted and inspected for both representative and extreme patients regardless of where the optimum falls. If the unconstrained optimisation consistently prefers $\alpha_0 < 1$, this is a model-misspecification signal; report both the unconstrained and constrained optima so the discrepancy can be investigated before finalising the model.

Prior-corrected centre. The predictive centre μ_n is a precision-weighted average of the population prior mean μ_0 and the patient's running mean μ_t. At $t = 1$ with $\kappa_0 = 1$ (for illustration; see Section 3.6 on why $\kappa_0 = 1$ is not the recommended choice), prior and observation receive equal weight. With a small fitted κ_0, the predictive centre converges to μ_t after even one or two observations. As $t \to \infty$, $\mu_n \to \mu_t$ regardless of κ_0. This shrinkage is appropriate when the patient's overlap history is short and the individual estimate is unreliable; the magnitude of the shrinkage is controlled by fitting κ_0 from the cohort data.

Connection to the branch update. The Welford update equations for (μ, σ) are algebraically identical to the NIG sufficient-statistics update for $(\bar{s}, M_2)$: both compute the same new mean and the same new sum of squared deviations. The belief-update operator $\mathcal{U}_{\mathrm{W}}$ is therefore already the correct NIG update and requires no modification in Stage B.

3.5 Stage B: Bellman Recursion

The Stage B recursion has the same state and structure as Stage A:

$$V_t^{\mathrm{B}}(o, D, \mu, \sigma) = \max_d \Big[-\hat{c}_t(o, d) + \mathbb{E}_{o' \sim \mathrm{St}_{2\alpha_n}(\mu_n,\, s_n)} V_{t+1}^{\mathrm{B}}\big(o', D + d,\ \mathcal{U}_{\mathrm{W}}(\mu, \sigma, o', t)\big) \Big], \tag{12}$$

where α_n, μ_n, and s_n are computed on-the-fly from (μ, σ, t) via equations (6)–(11), and $\mathcal{U}_{\mathrm{W}}$ is unchanged from Stage A.

3.6 Stage B: Implementation Strategy

Stage B requires three targeted changes to Stage A and no structural changes to the DP.

Change 1: Fraction-indexed branch-probability tables. In Stage A, `_P_BELIEF[mi, si, j]` is precomputed once at module load using Normal CDFs and is the same for all fractions. In Stage B, the branch probability depends on the fraction index t because the Student-t degrees of freedom $2\alpha_n(t)$ and predictive scale $s_n(\mu, \sigma, t)$ both depend on t. The precomputed table becomes:

$$\texttt{_P_BELIEF_NIG[t, mi, si, j]} \;=\; P_{\mathrm{St}}\Big(\text{bin } j \;\Big|\; \mu_n(\mu_{\mathrm{grid}}[m_i], t)\,,\ s_n(\mu_{\mathrm{grid}}[m_i], \sigma_{\mathrm{grid}}[s_i], t)\,,\ 2\alpha_n(t)\Big),$$

computed using Student-t CDF differences with the same left/right-tail folding as Stage A. The table shape is $(N_F, N_\mu, N_\sigma, N_o)$; for a 5-fraction treatment with the current grids this is $5 \times 150 \times 11 \times 441 \approx 3.6\,\mathrm{M}$ entries ($\approx 29\,\mathrm{MB}$), a negligible addition to module-load memory.

In the current codebase, `_P_BELIEF` is a module-level global accessed directly inside `_bellman_expectation_ful` (not passed as a parameter). `_P_BELIEF_NIG` must follow the same pattern: a module-level array of shape $(N_F, N_\mu, N_\sigma, N_o)$ precomputed at module load. The first axis is *0-based*: axis index 0 corresponds to $t = 1$ (fraction 1), axis index $t - 1$ corresponds to fraction t. In the function body, replace `_P_BELIEF[:, :, j]` with `_P_BELIEF_NIG[observation_count - 1, :, :, j]`. `_bellman_expectation` takes branch probabilities as an explicit argument and is unaffected by this change.

Variable treatment lengths. The public solver API accepts arbitrary `number_of_fractions`; the table must not be hard-wired to a single treatment length. At module load, precompute the table for `DEFAULT_NUMBER_OF_FRACTIONS` and cache the result in a module-level variable. If a caller requests a different treatment length, recompute for that length and update the cache. This is simpler than a fully general lazy-cache keyed by an arbitrary parameter and covers all current code paths, including the existing 3-fraction and 5-fraction test cases. Add regression tests exercising both.

Change 2: Planning-time initialisation. At decision time for fraction f, `current_overlap_probs` is currently computed by `probdist` (`helper_functions.py`), which returns Normal CDF differences without folding tail mass into the boundary bins; the function explicitly notes that its output does not sum to 1 if the distribution extends outside `_VOLUME_SPACE`. Under Stage B, this must be replaced with a Student-t predictive using parameters $(\mu_n, s_n, 2\alpha_n)$ evaluated at $n = f$ from the f actual observations. The replacement *must* fold left and right tails into the boundary bins, matching the behaviour of `_P_BELIEF_NIG`, so that planning-time and forward-DP branch probabilities are computed on the same probabilistic basis.
No σ point estimate required. The Student-t predictive parameters $(\mu_n, s_n, \nu_n = 2\alpha_n)$ follow directly from the NIG sufficient statistics $(n, \bar{s}, M_2)$ and the fitted hyperparameters via equations (6)–(11); the posterior does not need to be reduced to a scalar $\hat{\sigma}$ before the predictive can be evaluated. `std_calc` must not be called in the Stage B planning-time path. If a σ-grid index is needed (for example to look up an initial DP value), the NIG posterior mode of σ when $\sigma^2 \sim \text{InvGamma}(\alpha_n, \beta_n)$ is $\sqrt{\beta_n/(\alpha_n + \frac{1}{2})}$. Do not use $\sqrt{\beta_n/(\alpha_n - 1)}$, which equals $\sqrt{E[\sigma^2]}$ (the square root of the posterior mean of σ^2, not the mode); at small α_n this overestimates $\hat{\sigma}$ considerably.

Prior hyperparameters: refitting required. Stage A uses (α_0, β_0) (`DEFAULT_ALPHA`, `DEFAULT_BETA`) to regularise the MAP estimate of σ via a numerical grid search in `std_calc`. Critically, `std_calc` assumes a *Gamma*(α, β) prior on σ directly (density $\propto \sigma^{\alpha-1}\exp(-\sigma/\beta)$), whereas the NIG model assumes an *Inverse-Gamma*(α, β) prior on σ^2 (density $\propto (\sigma^2)^{-\alpha-1}\exp(-\beta/\sigma^2)$). These are materially different: with the current values, the Gamma prior on σ has mode $(\alpha_0 - 1)\beta_0 \approx 0.06$ cc, while the Inverse-Gamma prior on σ^2 with the same (α_0, β_0) numbers implies a σ-mode of $\sqrt{\beta_0/(\alpha_0 + 1)} \approx 0.61$ cc. Reusing `DEFAULT_ALPHA` and `DEFAULT_BETA` directly in equation (9) would silently produce incorrect β_n values.

Before Stage B is implemented, new constants `NIG_ALPHA_0` and `NIG_BETA_0` must be estimated from the patient cohort under the Inverse-Gamma convention. The recommended procedure is empirical Bayes: treat each patient's overlap sequence as i.i.d. draws from a Normal with unknown (μ, σ^2), place the NIG prior on (μ, σ^2), and maximise the marginal log-likelihood $\sum_i \log p(o_{i,1}, \ldots, o_{i,f_i} \mid \mu_0, \kappa_0, \alpha_0, \beta_0)$ over the full 58-patient cohort. Since κ_0 and μ_0 interact with α_0 and β_0 through the marginal (Student-t) likelihood, all four parameters should be fitted jointly with κ_0 included as a free parameter (bounds $[10^{-3}, 5]$). Do not fix $\kappa_0 = 1$: the 58-patient cohort has a wide spread of overlap means (0.01–21.18 cc), and $\kappa_0 = 1$ over-shrinks the predictive centre of extreme patients toward the population mean, which can represent a qualitative regression relative to Stage A at early fractions. After fitting, check the Stage B predictive centre at fraction 1 for the most extreme patients (patient 5, $\mu \approx 21$ cc; patient 3, $\sigma \approx 4.05$ cc) to confirm that shrinkage is clinically reasonable. Report the fitted κ_0 value prominently. The marginal likelihood for a single patient with observations $o_{1:n}$ and prior $(\mu_0, \kappa_0, \alpha_0, \beta_0)$ is a product of Student-t densities with sequentially updated parameters, which follows directly from applying equations (6)–(9) at each step.

`DEFAULT_ALPHA` and `DEFAULT_BETA` must be retained unchanged for `std_calc` and Stage A initialisation. The two additional constants μ_0 and κ_0 are also required and should be stored alongside `NIG_ALPHA_0` and `NIG_BETA_0` in `constants.py`.

Change 3: Planning-scan termination criterion. The current `precompute_plan` loop calls `get_state_space`, which defines the upper scanning bound from the 99.9th percentile of a Normal distribution fit to the current belief. Under Stage B this Gaussian heuristic must not be carried over unchanged. In the near-Cauchy regime ($\alpha_0^{\mathrm{NIG}} \approx 1$, small t), the Student-t 99.9th percentile can be an order of magnitude larger than the distribution's scale, producing an excessively slow scan or an ill-bounded upper limit. Replace the distribution-based criterion with a fixed clinically safe overlap bound combined with the existing policy-flat termination condition (policy fallen to the minimum deliverable dose). This decouples scan termination from the choice of predictive distribution and does not require re-validation when the predictive changes.

4 Required Numerical Design

4.1 State Discretisation

Stage B uses the same (μ, σ) grids and the same `_VOLUME_SPACE` as Stage A (Section 3.2). The belief-state resolution, value-function shape, and peak memory footprint are unchanged.

4.2 Complexity

Compared to Stage A, Stage B introduces:

- A fraction-indexed branch-probability table of $\approx 29\,\text{MB}$ at module load (versus $\approx 6\,\text{MB}$ for the Stage A table). This is a one-time cost per process invocation.
- Student-t CDF evaluation in place of Normal CDF during the precomputation step. `scipy.stats.t.cdf` is approximately 3–5$\times$ slower per call than `scipy.stats.norm.cdf`; the precomputation time grows from near-instant to a few seconds, but this cost is not incurred per patient.
- Per-node NIG parameter computation (five arithmetic operations per (m_i, s_i, t) triple) when calling `_bellman_expectation` with the planning-time Student-t distribution. This is negligible.

Per-patient DP runtime and peak memory are expected to remain within the Stage A budget ($\approx$ 35 s and $\approx$ 2.75 GB), because the per-patient Bellman loop is structurally unchanged and accesses an equally-sized branch-probability slice.

5 Validation and Acceptance Criteria

5.1 Implementation Correctness Gates

1. Mathematical consistency: NIG posterior updates, predictive distribution, and branch-probability tables in code match this document.

2. Functional correctness: full test suite passes, and new tests cover fraction-dependent branch probabilities, NIG initialisation, and the equivalence of the Welford update with the NIG sufficient-statistics update.

3. Baseline lock-in: Stage A benchmark artifacts (58-patient cohort replay) are frozen before Stage B development begins and serve as the exclusive reference for all Stage B comparisons.

4. Reproducibility: deterministic outputs under fixed discretisation and random settings.

5.2 Evaluation Metrics

For each trajectory i on the fixed 58-patient replay cohort, define:

- B_i^{A}: Stage A benefit,
- B_i^{B}: Stage B benefit,
- B_i^{ideal}: best-case upper-bound benefit.

All three are measured on the same trajectory under identical simulation conditions.

Primary per-trajectory improvement:

$$\Delta_i = B_i^{\mathrm{B}} - B_i^{\mathrm{A}}. \tag{13}$$

Distance-to-ideal:

$$G_i^{\mathrm{A}} = B_i^{\mathrm{ideal}} - B_i^{\mathrm{A}}, \qquad G_i^{\mathrm{B}} = B_i^{\mathrm{ideal}} - B_i^{\mathrm{B}}. \tag{14}$$

Normalised gap closure:

$$R_i = \frac{B_i^{\mathrm{B}} - B_i^{\mathrm{A}}}{\max\bigl(B_i^{\mathrm{ideal}} - B_i^{\mathrm{A}},\ \epsilon\bigr)}, \tag{15}$$

with small $\epsilon > 0$ for numerical safety. $R_i = 0$ means no progress over Stage A; $R_i = 1$ means the full Stage A-to-ideal gap is closed.

5.3 Acceptance Criteria

1. Mean improvement is positive: $\mathbb{E}[\Delta_i] > 0$ with a positive paired-bootstrap 95% confidence-interval lower bound.

2. Gap-to-ideal shrinks on average: $\mathbb{E}[G_i^{\mathrm{B}}] < \mathbb{E}[G_i^{\mathrm{A}}]$ and median $R_i > 0$.

3. Tail safety is not degraded: the fraction of cases with $\Delta_i < -1$ ccGy does not increase relative to Stage A.

4. Runtime and memory remain within the Stage A budget.

6 Stage B Evaluation Protocol

Stage B must be evaluated in the following order before any model-selection or simplification decisions are made:

1. Run the full 58-patient replay benchmark under both Stage A and Stage B from the same frozen artifacts.
2. If the mean improvement is negligible ($|\mathbb{E}[\Delta_i]| < 0.01\,\text{ccGy}$): record this as a validated negative result. Stage A is accepted as the production model and Stage B is not promoted.
3. If the improvement is meaningful: compare in detail across:
   (a) expected total penalty and mean Δ_i,
   (b) tail penalty at p90 and p95,
   (c) per-fraction dose trajectory differences,
   (d) terminal feasibility (underdose and overdose incidence),
   (e) runtime and peak memory.
4. Validate generalisation via leave-one-out cross-validation: for each patient, refit the NIG hyperparameters on the remaining 57 patients and evaluate on the held-out patient. Compare the LOO $\mathbb{E}[\Delta_i]$ with the in-sample estimate from step 1. If the two are within noise, proceed with the full-cohort fit for the final model. If they diverge, investigate whether the hyperparameters are absorbing cohort-specific structure before promoting Stage B. This step applies regardless of whether the improvement in steps 2–3 is negligible or meaningful. Leave-one-out is straightforward to implement because the marginal likelihood factorises over patients and each patient's contribution can be excluded by omitting it from the sum.

Any subsequent simplification of Stage B — for example reverting to a Normal predictive for fractions $t \geq t^*$ where the Student-t and Normal are empirically indistinguishable — must pass the same comparison protocol against the full Stage B result before being accepted.